

Linear Algebra for Political Scientists

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Structure

1. Regression Notation, At Long Last
2. Least Squares, Geometrically
3. The Normal Equations
4. Variance-Covariance Matrices
5. Standard Errors & Collinearity

The Need for New Notation

Up until now: $A\mathbf{x} = \mathbf{b}$, where $A \in \mathbb{R}^{n \times n}$

Recall our Estonia/Latvia example:

$$\begin{bmatrix} 7 & 9 \\ 6 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$$

...or, really, any of the Gaussian elimination examples from Session 2:

$$x + y + z = 6$$

$$2x + y + 3z = 13$$

$$x - y + z = 2$$

“Real” Systems in Political Science

X is not square!
 6×2

GDP Growth	EU Integration
3	2
4	1
3	3
7	9
6	8
7	8

X

$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ β

$=$

Democracy Score
3
2
4
8
7
8

y

$$A\mathbf{x} = \mathbf{b} \quad \text{Old Notation}$$

$$X\beta = y \quad \text{New Notation}$$

the vector of unknown weights
(for which we must solve)

Overdetermined System

$$3w_1 + 2w_2 = 3$$

$$4w_1 + w_2 = 2$$

$$3w_1 + 3w_2 = 4$$

$$7w_1 + 9w_2 = 8$$

$$6w_1 + 8w_2 = 7$$

$$7w_1 + 8w_2 = 8$$

A system is **overdetermined** when it has more equations than unknowns.

Any **two** of these six equations allow us to solve for w_1, w_2 uniquely (this would be an ordinary square system). The problem is the other four equations.

Once β is fixed by, say, the first two countries, there is no reason at all the remaining four equations should also hold. They involve the **same two unknowns**, already spoken for, with no freedom left to satisfy anything new.

Reframing Our Goal

Old Notation $A\mathbf{x} = \mathbf{b}$

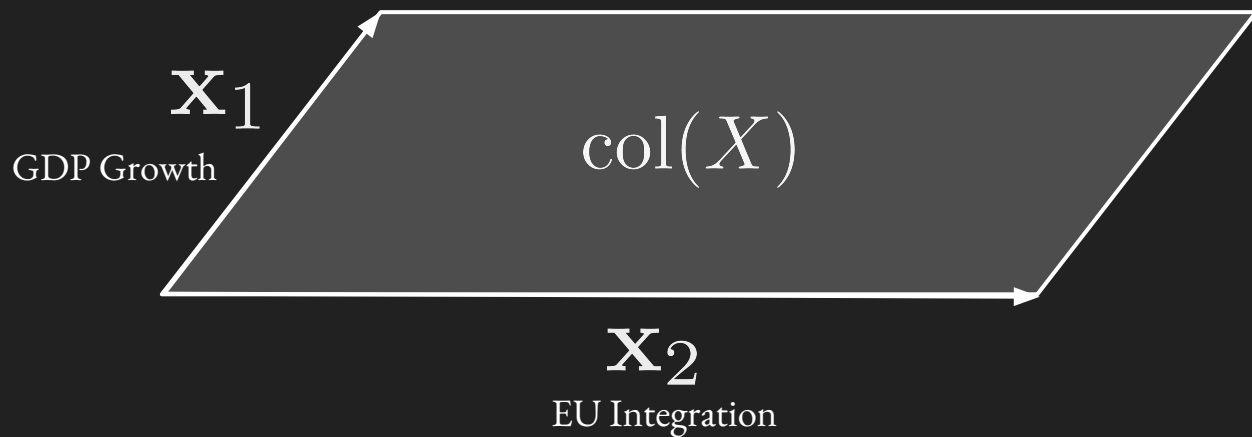
In this case, we can solve for $\mathbf{x}$ exactly via Gaussian elimination, since A is square.

New Notation $X\beta = y$

Here, we **cannot** solve for β exactly, since X is **not square**. So instead, we will solve for the β that gets $X\beta$ as close as possible to y .

The Column Space of a Matrix

The column space of a matrix X is the set of all vectors that can be written as $X\beta$ for some choice of β . Equivalently, it is every linear combination of X 's columns.



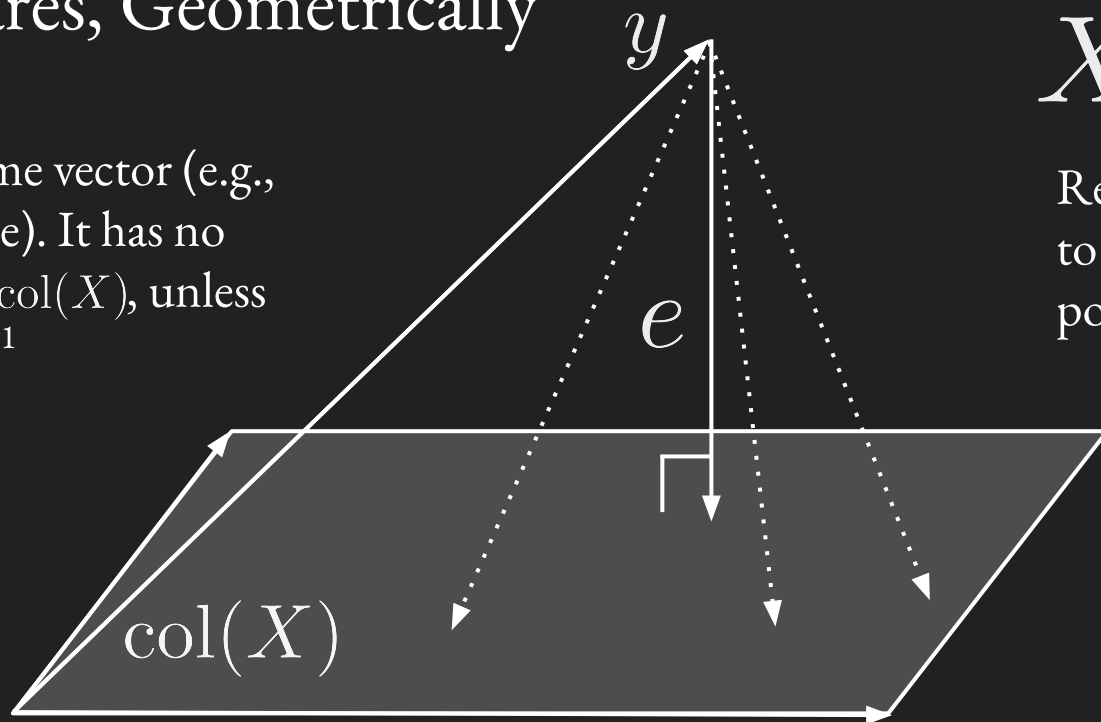
$$\text{col}(X) = \{X\beta : \beta \in \mathbb{R}^k\}$$

Least Squares, Geometrically

y is our outcome vector (e.g., democracy score). It has no reason to be in $\text{col}(X)$, unless by coincidence.¹

$$X\beta = y$$

Recall: We want $X\beta$ to be as close as possible to y .

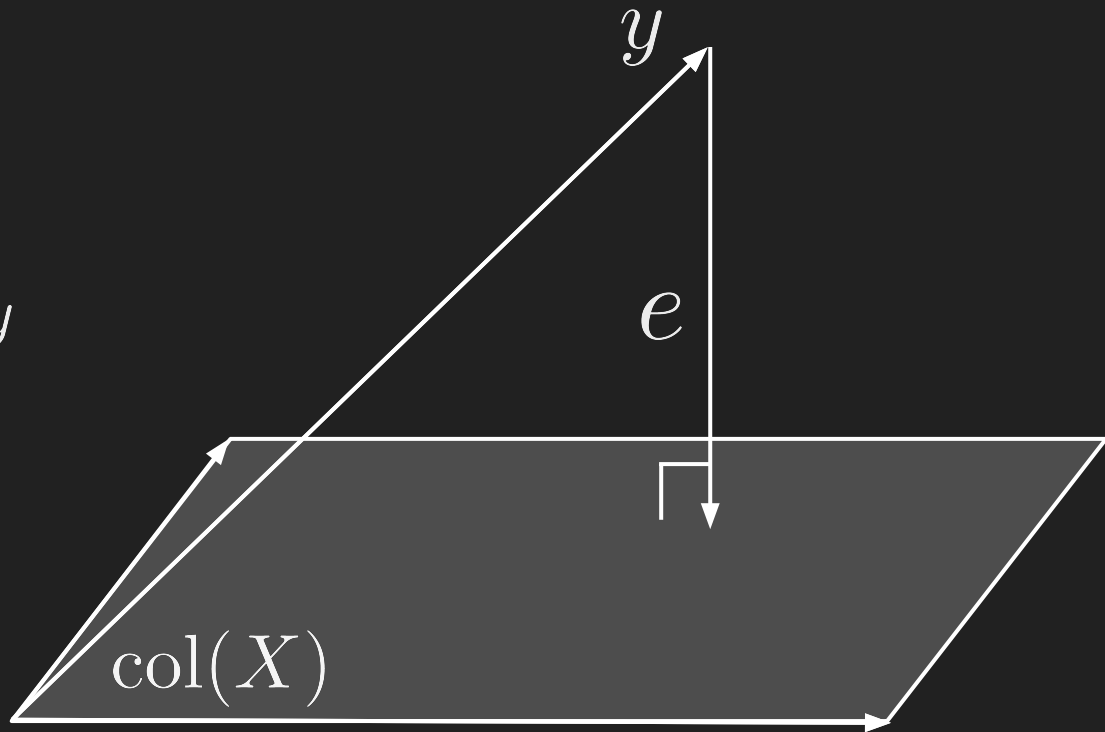


$$\text{col}(X) = \{X\beta : \beta \in \mathbb{R}^k\}$$

¹ If democracy score happened to be a linear combination of GDP growth and EU integration.

Orthogonal Projection (Least Squares)

The least-squares solution $\hat{\beta}$ is the value of β that minimizes the length of the residual, $\|y - X\beta\|$. Geometrically, $X\hat{\beta}$ is the orthogonal projection of y onto the column space of X : the closest point in that space to y , found by dropping a perpendicular line.



Fitted Values

$X\hat{\beta}$ itself is called $\hat{y}$: the vector of *fitted* (or *predicted*) values, as opposed to y , the vector of actually observed outcomes:

$$\hat{y} = X\hat{\beta}$$

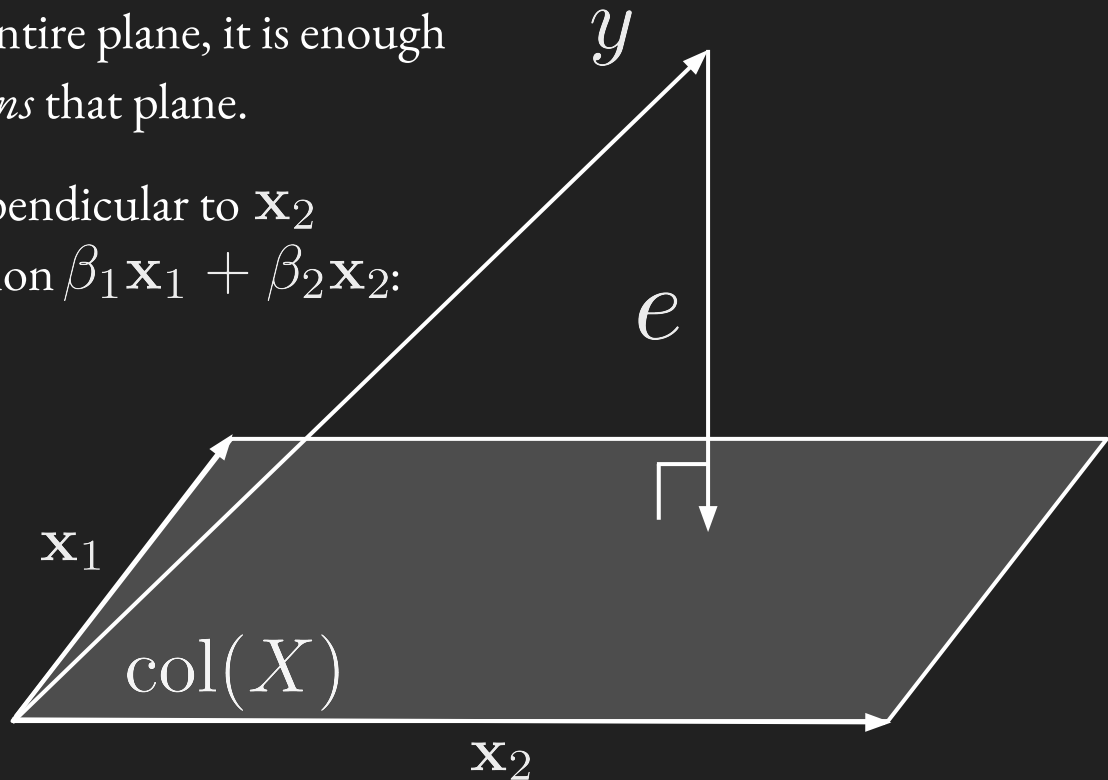
$\hat{y}$ and $X\hat{\beta}$ are the same object; $\hat{y}$ is simply the lighter notation. The residual is, correspondingly, $e = y - \hat{y}$.

Perpendicular to a Plane?

In order to be perpendicular to an entire plane, it is enough to be perpendicular to whatever *spans* that plane.

If e is perpendicular to $\mathbf{x}_1$ and perpendicular to $\mathbf{x}_2$ individually, then for *any* combination $\beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2$:

$$\begin{aligned} & e \cdot (\beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2) \\ &= \beta_1 (e \cdot \mathbf{x}_1) + \beta_2 (e \cdot \mathbf{x}_2) \\ &= \beta_1 (0) + \beta_2 (0) = 0 \end{aligned}$$



Setting up the Normal Equations

Let $\mathbf{x}_1, \mathbf{x}_2$ be the two columns of X (growth and integration). For e to be perpendicular to $\text{col}(X)$, it must be perpendicular to $\mathbf{x}_1$ and $\mathbf{x}_2$.

$$\mathbf{x}_1 \cdot e = 0, \quad \mathbf{x}_2 \cdot e = 0$$

(Recall that two vectors are orthogonal if and only if their dot product is zero).

These two conditions are equivalent to:

$$X^\top e = 0$$

The Normal Equations & OLS Estimator

Substituting $e = y - X\hat{\beta}$ into $X^\top e = 0$, we get:

Normal Equations

$$X^\top (y - X\hat{\beta}) = 0 \quad \implies \quad \boxed{X^\top X \hat{\beta} = X^\top y}$$

Pre-multiplying both sides by $(X^\top X)^{-1}$ (assuming $X^\top X$ is invertible), we get:

$$\boxed{\hat{\beta} = (X^\top X)^{-1} X^\top y} \longrightarrow \text{OLS Estimator}$$

A Numerical Example

$$X^{\top} X = \begin{bmatrix} 3 & 3 & 6 \\ 2 & 3 & 8 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 3 & 3 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 54 & 63 \\ 63 & 77 \end{bmatrix}$$

$$(X^{\top} X)^{-1} = \begin{bmatrix} 54 & 63 \\ 63 & 77 \end{bmatrix}^{-1}$$

$$X^{\top} y = \begin{bmatrix} 3 & 3 & 6 \\ 2 & 3 & 8 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 63 \\ 74 \end{bmatrix}$$

$$\implies \hat{\beta} = \begin{bmatrix} 1 \\ \frac{1}{7} \end{bmatrix}$$

$$X = \begin{bmatrix} 3 & 2 \\ 3 & 3 \\ 6 & 8 \end{bmatrix} \begin{array}{l} \text{Armenia} \\ \text{Georgia} \\ \text{Latvia} \end{array}$$

GDP EU
Growth Integration

$$y = \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix}$$

Democracy
Score

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

A Numerical Example: Solving for Fitted Values

$$\hat{\beta} = \begin{bmatrix} 1 \\ \frac{1}{7} \end{bmatrix} \rightarrow \text{Our vector of weights} \\ \text{(GDP growth is basically} \\ \text{doing all the work here).}$$

$$\hat{y} = X\hat{\beta} = \begin{bmatrix} 3(1) + 2(\frac{1}{7}) \\ 3(1) + 3(\frac{1}{7}) \\ 6(1) + 8(\frac{1}{7}) \end{bmatrix} = \begin{bmatrix} \frac{23}{7} \\ \frac{24}{7} \\ \frac{50}{7} \end{bmatrix} \quad \dots \text{compared to} \dots \quad y = \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix}$$

So how “off” are our estimates? We can answer this question by finding the residuals.

A Numerical Example: Finding the Residuals

$$e = y - \hat{y} = \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix} - \begin{bmatrix} \frac{23}{7} \\ \frac{24}{7} \\ \frac{50}{7} \end{bmatrix} = \begin{bmatrix} -\frac{2}{7} \\ \frac{4}{7} \\ -\frac{1}{7} \end{bmatrix}$$

Check to make sure that the residual is orthogonal to both columns of $X = \begin{bmatrix} 3 & 2 \\ 3 & 3 \\ 6 & 8 \end{bmatrix}$:

$$\mathbf{x}_1 \cdot e = 3 \left(-\frac{2}{7}\right) + 3 \left(\frac{4}{7}\right) + 6 \left(-\frac{1}{7}\right) = -\frac{6}{7} + \frac{12}{7} - \frac{6}{7} = 0$$

$$\mathbf{x}_2 \cdot e = 2 \left(-\frac{2}{7}\right) + 3 \left(\frac{4}{7}\right) + 8 \left(-\frac{1}{7}\right) = -\frac{4}{7} + \frac{12}{7} - \frac{8}{7} = 0$$

So the residual is indeed orthogonal to the column space of X , as expected.

Variance-Covariance Matrices: Motivation

How spread out is a variable? How do two variables move together?

Recall the sample variance formula: $\text{Var}(x) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

Let $\tilde{\mathbf{x}}$ be the **centered** version of x (mean subtracted from every entry, $\tilde{x}_i = x_i - \bar{x}$):

$$\tilde{\mathbf{x}} \cdot \tilde{\mathbf{x}} = \sum_{i=1}^n \tilde{x}_i^2 = \sum_{i=1}^n (x_i - \bar{x})^2$$

Variance is nothing more than a centered column dotted with itself, divided by $n - 1$.

Identical logic for covariance:

$$\tilde{\mathbf{x}} \cdot \tilde{\mathbf{y}} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$$

Variance-Covariance Matrices

Suppose we have k variables, each centered: $\tilde{\mathbf{z}}_1, \dots, \tilde{\mathbf{z}}_k$. The variance-covariance matrix Σ is the $k \times k$ matrix whose (i, j) entry is exactly the computation from the previous slide, applied to variables i and j :

$$\Sigma_{ij} = \frac{1}{n-1} \tilde{\mathbf{z}}_i \cdot \tilde{\mathbf{z}}_j$$

When $i = j$, this is variable i 's own variance; when $i \neq j$, it is the covariance between variables i and j . If $\tilde{\mathbf{Z}}$ is the matrix whose columns are $\tilde{\mathbf{z}}_1, \dots, \tilde{\mathbf{z}}_k$, then all of these dot products are computed at once by

$$\Sigma = \frac{1}{n-1} \tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}}$$

Variance-Covariance Matrices: Example

For growth and integration alone, Σ is a 2×2 matrix:

$$\begin{array}{cc}
 \begin{bmatrix} 3 & 2 \\ 4 & 1 \\ 3 & 3 \\ 7 & 9 \\ 6 & 8 \\ 7 & 8 \end{bmatrix} & \Sigma = \begin{bmatrix} \text{Var}(\text{growth}) & \text{Cov}(\text{growth, integration}) \\ \text{Cov}(\text{integration, growth}) & \text{Var}(\text{integration}) \end{bmatrix} \\
 \text{GDP} & \text{EU} \\
 \text{Growth} & \text{Integration}
 \end{array}
 \quad \parallel \quad
 \begin{bmatrix} \frac{18}{5} & \frac{31}{5} \\ \frac{31}{5} & \frac{377}{30} \end{bmatrix}
 \approx
 \begin{bmatrix} 3.6 & 6.2 \\ 6.2 & 12.57 \end{bmatrix}$$

Zooming Out: The True Model

$$y = X(\beta) + (\varepsilon)$$

The true, unknown relationship connecting growth and integration to democracy.

Everything else left over (measurement error, omitted variables, etc.).

We estimate this using $\hat{\beta}$, which depends on the specific **sample** of countries. With a different sample, $\hat{\beta}$ will be different! So we need to know how $\hat{\beta}$ **varies**.

We will assume that each entry of ε has some fixed, but unknown variance σ^2 .

This is a variance-covariance question.

$\hat{\beta}$ versus β

We previously derived $\hat{\beta} = (X^\top X)^{-1} X^\top y$.

Substituting the true model $y = X\beta + \varepsilon$ into the above expression, we get:

$$\begin{aligned}\hat{\beta} &= (X^\top X)^{-1} X^\top y = (X^\top X)^{-1} X^\top (X\beta + \varepsilon) \\ &= \beta + (X^\top X)^{-1} X^\top \varepsilon\end{aligned}$$

So $\hat{\beta}$ differs from β by exactly $(X^\top X)^{-1} X^\top \varepsilon$: a fixed matrix multiplied by the random error term (we will use this fact in a moment).

Variance of a Linear Transformation

Recall that for a scalar a and a random variable X , $\text{Var}(aX) = a^2 \text{Var}(X)$.

Similarly, for a matrix A and a random vector $\mathbf{z}$ with variance-covariance matrix Σ ,

$$\text{Var}(A\mathbf{z}) = A\Sigma A^\top$$

Next Step: Use this definition to derive the variance of the OLS estimator, $\hat{\beta}$.

Variance of the OLS Estimator

Recall that $\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon$.

Let $A = (X^\top X)^{-1} X^\top$, so $\hat{\beta} - \beta = A\varepsilon$. Then,

$$\text{Var}(\hat{\beta}) = A(\sigma^2 I)A^\top = \sigma^2 AA^\top$$

Computing $AA^\top$ directly, we get:

$$\begin{aligned} AA^\top &= (X^\top X)^{-1} X^\top \left(X (X^\top X)^{-1} \right) \\ &= (X^\top X)^{-1} (X^\top X) (X^\top X)^{-1} = (X^\top X)^{-1} \end{aligned}$$

Therefore, $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$.

Standard Errors

The standard error of a coefficient $\hat{\beta}_j$ is the square root of the corresponding diagonal entry of $\text{Var}(\hat{\beta})$:

$$\text{SE}(\hat{\beta}_j) = \sqrt{\sigma^2 [(X^\top X)^{-1}]_{jj}}$$

For example, if $\hat{\beta} = \begin{bmatrix} \hat{w}_1 \\ \hat{w}_2 \end{bmatrix}$, then $\text{Var}(\hat{\beta}) = \begin{bmatrix} \text{Var}(\hat{w}_1) & \text{Cov}(\hat{w}_1, \hat{w}_2) \\ \text{Cov}(\hat{w}_2, \hat{w}_1) & \text{Var}(\hat{w}_2) \end{bmatrix}$.

Then, $\text{SE}(\hat{w}_1) = \sqrt{\text{Var}(\hat{w}_1)} = \sqrt{\sigma^2 [(X^\top X)^{-1}]_{11}}$

From $\text{Var}(\hat{\beta})$ to the Regression Table

Consider our previous example involving Armenia, Georgia, and Latvia. Let's regress Democracy Score on GDP Growth and EU Integration and build the variance-covariance matrix:

$$\text{Var}(\hat{\beta}) = \begin{bmatrix} \text{Var}(\hat{w}_1) & \text{Cov}(\hat{w}_1, \hat{w}_2) \\ \text{Cov}(\hat{w}_2, \hat{w}_1) & \text{Var}(\hat{w}_2) \end{bmatrix} = \begin{bmatrix} 0.175 & -0.143 \\ -0.143 & 0.122 \end{bmatrix}$$

Predictor	Coefficient	
GDP Growth	1.000	
	(0.418)	$\longrightarrow \sqrt{0.175}$
EU Integration	0.143	
	(0.350)	$\longrightarrow \sqrt{0.122}$

Standard Errors & Collinearity

$$\text{Var}(\hat{\beta}) = \sigma^2 \boxed{(X^\top X)^{-1}}$$



If X is rank deficient, then so is $X^\top X$ ($\text{rank}(X) = \text{rank}(X^\top X)$).
If the columns of $X^\top X$ are highly collinear, its determinant will be near 0 and its entries will blow up!



If the entries of $(X^\top X)^{-1}$ are large, the entries of $\text{Var}(\hat{\beta})$ are too. So collinearity results in large standard errors.

$$A\mathbf{x} = \mathbf{b}$$

square system

$$X\beta = y$$

overdetermined system

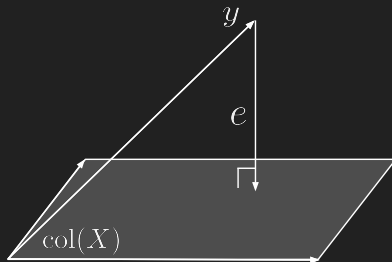
$$\text{col}(X)$$

every possible $X\beta$

Standard Errors
square root of diagonal

$$\text{Var}(\hat{\beta})$$
$$\sigma^2(X^\top X)^{-1}$$

Collinearity
 $(X^\top X)^{-1}$ blows up



Orthogonal Projection

$$\text{Normal Equations}$$
$$X^\top X \hat{\beta} = X^\top y$$

$$\text{OLS Estimator}$$
$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$